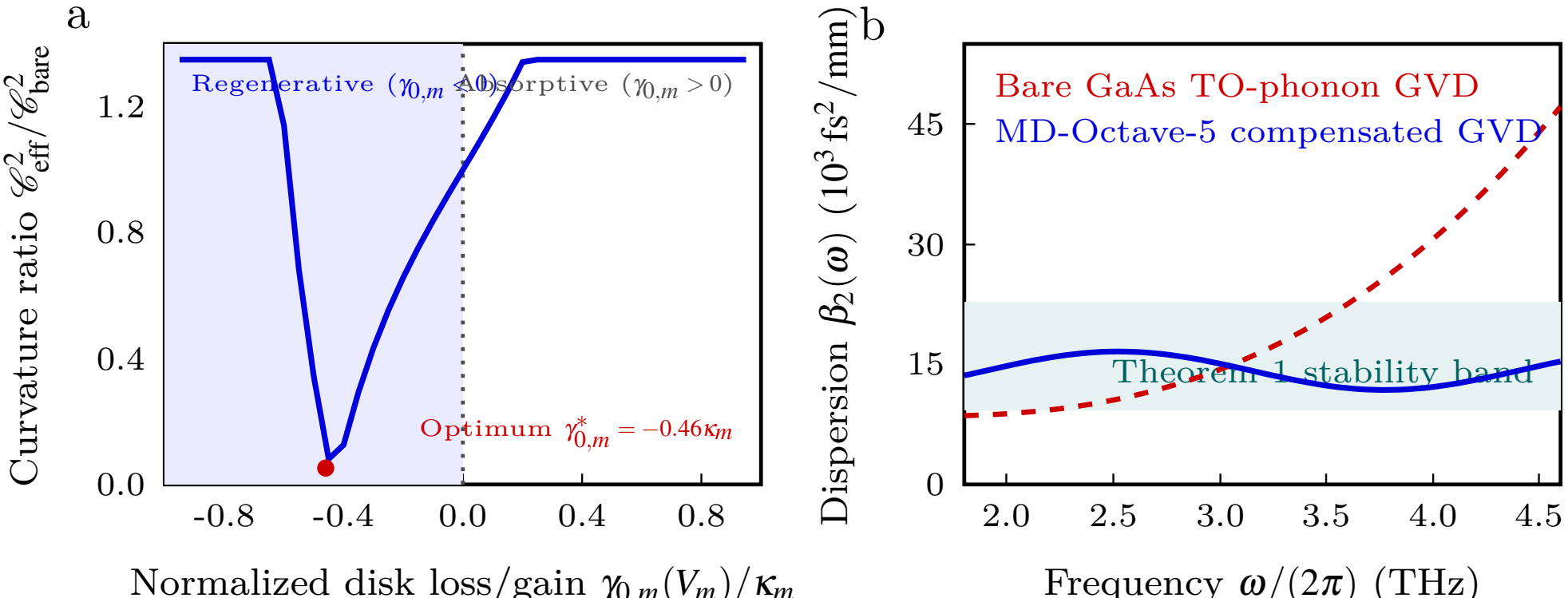
a
Regenerative ($\gamma_{0,m} < 0$)
Absorptive ($\gamma_{0,m} > 0$)
Optimum $\gamma^*_{0,m} = -0.46\kappa_m$
Curvature ratio $\mathscr{C}^2_{\rm eff}/\mathscr{C}^2_{\rm bare}$
1.2
0.8
0.4
0.0
-0.8
-0.4
0.0
0.4
0.8
Normalized disk loss/gain $\gamma_{0,m}(V_m)/\kappa_m$
b
Bare GaAs TO-phonon GVD
MD-Octave-5 compensated GVD
Theorem 1 stability band
Dispersion $\beta_2(\omega)$ (10^3 fs^2/mm)
45
30
15
0
2.0
2.5
3.0
3.5
4.0
4.5
Frequency $\omega/(2\pi)$ (THz)